



**VIT**  
**CHENNAI**



# **NATURAL LANGUAGE PROCESSING**

## **REVIEW - 3**

**PROJECT TITLE: AI RESUME MATCHER**

**COURSE CODE: SWE1017**

**SLOT: F2**

**FACULTY: DR.B. SALEENA**

**BY,**

**22MIA1118 – P SAI ESWARI**

**22MIA1146 – TEJASH KUMAR G S**

## **TABLE OF CONTENTS:**

| Section                         | Page(s) |
|---------------------------------|---------|
| Abstract                        | 3       |
| Introduction                    | 4       |
| Problem Definition & Objectives | 5       |
| Dataset & Preprocessing         | 6-7     |
| Algorithms Used                 | 8-12    |
| Implementation Details          | 12-13   |
| Results & Discussion            | 13-14   |
| Conclusion                      | 14-15   |
| References                      | 15-16   |

# **1. ABSTRACT:**

Recruiting can be one of the most burdensome and expensive activities in HR management. Traditional keyword based algorithmic matching techniques do not capture the true semantics of the candidate profiles or the job description, motivating the need for automatic text and semantic analysis, recognizing the candidate's elements that express their contextualized suitability with the job description. This research proposes an AI-Based Resume Matcher focusing on Natural Language Processing (NLP) capabilities to produce a matching of contextualized representations of candidate's resumes and the job descriptions, using a natural matching process based on semantic content rather than traditional keyword matches.

The AI-Based Resume Matcher applies Sentence-BERT [SBERT] to produce sentence embeddings to represent each candidate's resume and job description at the sentence level, before comparing the resume and job description embeddings and calculating cosine similarity on the resumes of matched candidates, producing a ranked list of matched applicants. The model also incorporates an unintended consequence whereby the engine highlights parts of the resume that were most relevant to the matching job description to provide an element of interpretability for recruiters. Additionally, the model can support a number of different file formats including PDF, DOCX, and TXT formats and has been implemented with Flask for the user interface.

In summary, the proposed model will contribute to improving the recruitment screening process by automating the candidate screening process, improving the accuracy of candidate-job fit predictions on candidates who matched the job description, and improve transparency through an explainable AI components.

## **2. INTRODUCTION**

With job applicants multiplying overnight and resumes flooding the internet, companies are grappling with the manual effort to process thousands of applications and resumes for each job opening. Traditional Applicant Tracking Systems (ATS) rely on keyword matching; resumes are ranked based on their matching frequency to the keywords in the job description. While this simple filtering solution works well for the simplest job descriptions, it fails to rank resumes with qualified candidates because they use different wording or phrasing for the same idea.

Recent advances in Natural Language Processing (NLP) and deep learning have allowed the ability of computers to semantically understand meaning and context, making them a good option for comparing documents, ranking documents, etc. Models like BERT (Bidirectional Encoder Representations from Transformers) and its variants like Sentence-BERT (SBERT) can both leverage contextual relationships between words and sentences, leading to a stronger comprehension of text.

The AI-Based Resume Matcher utilizes the matched learning of uncertainty with NLP, to create a contextual resume screening system that compares resumes to the job description on the semantic similarity of their meaning. Instead of counting keywords, it identifies how meaningfully and qualified a candidate is for the job based on their experience. The model is also interpretable because it highlights relevant sentences counted from the resumes that justified why a candidate received their score. In the end, this system allows recruiters to:

- Save time on the screening side
- Improve accuracy when candidates match with jobs
- Gain more transparent insights through snippet highlighting
- Scale to large datasets efficiently through automation

### **3. PROBLEM DEFINITION & OBJECTIVES**

#### **3.1 PROBLEM DEFINITION**

Recruiters often spend hours meticulously examining resumes to find qualified candidates. Traditional keyword-based matching generally cannot determine the semantic relevance between the job description (JD) and a candidate's resume. The aim of this project is to automatically screen resumes using state-of-the-art NLP and ranking resumes semantically based on their relevance to a given JD and snippets of the top-most relevant sentences. The project seeks to enhance accuracy, lessen burden on recruiters, and help provide contextually pursuing relevant resumes, even without matching keywords.

#### **3.2 OBJECTIVES**

The main goals of the projects is to:

- Implement an intelligent resume ranking to compute the semantic similarity between resumes and job descriptions utilizing NLP techniques and deep learning
- apply the Sentence-BERT embeddings to faithfully imitate the contextual meaning of textual data.
- utilize cosine similarity to assess a relevance score and ranking resumes;
- Implement the relevant text fragments from resumes, as the model is more understandable and reliable.
- Web-based app with resumes input and ranking via Flask or any other frameworks.
- The possibility to scale and input resumes in any format.pdfs,.docx,.txt
- Evaluate performance through Top-k accuracy and human validation

## **4. DATASET & PREPROCESSING**

### **4.1 DATASET DESCRIPTION**

#### **Dataset Type:**

Customized set of resumes (PDF, DOCX, TXT) plus job descriptions (plain text.)

#### **Source:**

Can be collected from anonymized resumes made publicly available (e.g., Kaggle resume sets) or simulated for testing.

#### **Attributes:**

- Candidate Name (optional)
- Resume Text (the main body of the content)
- Skills, Experience, Education (embedded in the text as appropriate)
- Job Description (inputted by recruiting department.)

#### **Size:**

Ranges typically from 50-500 resumes depending on what the organization would like to test, basing the resumes against individual job descriptions.

### **4.2 PREPROCESSING STEPS**

Resumes and job descriptions are essentially unstructured text data. Therefore, serious preprocessing should be conducted to form quality embeddings and be able to compute accurate similarities. Namely, the following steps should be implemented:

**File Parsing:**

Perform extraction of textual content from PDF, combining the PyPDF2 library. Further, combine the docx2txt library to perform extraction of.docx content. For.txt files, it is read directly.

**Lowercasing:**

Make all text lowercase to provide equal processing.

**Noise Removal:**

Perform removal of special characters, numbers, hyperlinks and drop-pointless symbols.

**Segment Sentence:**

Tokenization of sentences splits text into sentences, which is used in snippet-level analysis using `nltk.sent_tokenize()`.

**Stopword Removal:**

Stopword removal eliminates common words such as “is”, “and”, “the”, etc. stopword removal helps to normalize the text.

**Lemmatization:**

Lemmatization is the process of reducing words to their base or dictionary form. Lemmatization is useful for better semantic consistency in text.

**Vectorization:**

The textual data is transformed using Sentence BERT sentence embeddings into dense vectors which are used in compute similarity.

## **5. ALGORITHMS USED**

### **5.1 Workflow Overview**

1. Input Job Description
2. Upload Resumes (PDF/DOCX/TXT)
3. Text Preprocessing
4. Sentence Embedding using Sentence-BERT
5. Compute Cosine Similarity
6. Rank Resumes by Relevance
7. Highlight Top Matching Sentences
8. Display Results in Web Interface

### **5.2 ALGORITHMS:**

#### **Algorithm 1: Sentence-BERT (SBERT)**

**Concept:** SBERT is just a BERT with the goal of computing semantically meaningful sentence embeddings.

#### **Working Principle:**

- Converts both resume and job description to high-dimensional vectors (embeddings).
- These embeddings represent contextual meaning, rather than merely word overlap.
- Allows you to semantically compare two text documents.

**Reason for Use:** Captures context, synonyms, job-relevant semantics (“data analyst” and “data scientist” as being related), etc.

**Implementation Library:** sentence-transformers

**Desired output:** More accurate results and better contextual understanding in resume ranking.

### **Algorithm 2: Cosine Similarity**

**Idea:** Cosine similarity is computed between two embedding vectors to judge if their meaning are close.

**Formula:**

$$\text{Cosine Similarity} = \frac{A \cdot B}{||A|| \times ||B||}$$

where  $A$  and  $B$  are vector representations of job description and resume text.

**Use Case:**

- Used after SBERT embedding generation.
- Returns a similarity score ranging between 0 (no match) and 1 (perfectly matching).

**Outcome:**

Enables scoring the resumes on semantical relevance with JD.

### **Algorithm 3: TF-IDF as in (Term Frequency-Inverse Document Frequency)**

**Idea :** A statistical measure of how important a word is in document, compared to a collection (+tf-idf).

**Role in Project:**

- Fallback in environments without deep-learning models.
- Treats resumes and JD as sparse vectors using cosine similarity.

**Benefit:**

- Simple and computationally light.
- Serves as a reference point against which we compare the performance of SBERT.

**Algorithm 4: A Sentence Snippet Extraction by Local Similarity****Concept:**

- Splits resumes into sentences.
- Calculates cosine similarity distances between each sentence and the job description's embedding.
- Ranking best top-k sentences with high-level similarity as “highlighted snippets”.

**Use Case:**

Provides HRs an instant impression of why the candidate is good for JD.

**Outcome:**

Improves interpretability of results.

**5.3 INNOVATIONS:**

- **Semantic Resume Ranking:** Instead of simply matching keywords, context-aware comparison is performed with transformer-based embeddings from Sentence-BERT.
- **Sentence Snippet Highlighting:** Shows a sentence snippet atop each resume for interpretability.

- **Hybrid Model (SBERT + TF-IDF) :** This model is a blend between deep learning and classical NLP that gives reliable performances in any context.
- **Multi-format Resume Supported:** Handles resumes in PDF, DOCX and TXTfile format automatically.
- **Contextual Text Cleaning:** Preserves semantically rich words and discards irrelevant clause content for faithful embedding.
- **Web Friendly Integration:** Interactive resume upload and ranking dashboard using flask.
- **Scalable Architecture:** The model is lightweight and can process large batches of resumes in very small time.
- **Transparency:** Boils down what mattered in the content to influence rank, so the AI is understandable and usable.

## **5.4 EVALUATION METRICS:**

- **Cosine Similarity Score:** It is the main metric to measure similarity between job description and resumes.
- **Top-K Accuracy:** Checks whether the top K results have best-matching resumes.
- **Precision & Recall:** Applied when labeled “fit” vs. “not-fit” data can be utilized to validate performance.
- **Mean Reciprocal Rank (MRR):** It measures the relevance of the ranking how soon does correct resume rank high.
- **Human Validation:** Highlights are viewed as snippets for qualitative evaluation by recruiters.
- **Lead Time:** Monitors the average time per resume in order to establish system scalability.

- **Comparative Analysis:** Comparison between SBERT models and TF-IDF baseline.

## 6. IMPLEMENTATION DETAILS

### 6.1 TOOLS AND LIBRARIES

| Component            | Tool / Library                        |
|----------------------|---------------------------------------|
| Programming Language | Python                                |
| Web Framework        | Flask                                 |
| NLP Model            | Sentence-BERT (sentence-transformers) |
| Text Processing      | NLTK, regex                           |
| File Parsing         | PyPDF2, docx2txt                      |
| ML Utilities         | Scikit-learn, NumPy, Pandas           |
| Frontend             | HTML, CSS, JavaScript                 |

### 6.2 SYSTEM ARCHITECTURE

#### **Frontend:**

The frontend system is used by a recruiter to upload resumes and enter job descriptions.

**Backend:** The backend will manage all text extraction, preprocessing generation of embeddings, and computation of similarities.

**Model Server:** The Sentence-BERT model is executed by the embedding generator

**Output Layer:** The generated resume are ranked and relevant snippets are then highlighted

## **6.3 EXECUTION FLOW**

1. Recruiter uploads job description as well as resumes
2. Text data is prepared by the system
3. The Sentence-BERT model generates vector embeddings
4. A similarity score is computed using cosine distance for each resume
5. Resumes are then ranked in a descending order based on the similarity score

The top-k resumes are shown, along with the highlighted relevant sentences.

## **7. RESULTS & DISCUSSION**

### **7.1 QUANTITATIVE RESULTS**

- The system successfully matched and ranked the uploaded resumes with the provided job description based on the semantic similarity factor.
- The best ranked resume was data-analyst2 with a similarity score of 38.51% which means strong context matching between the candidate experience and job requirements.
- Followed by 28.74%, 28.36%, and 17.07% score resumes which are moderately to low on semantic alignment.
- The model could differentiate between highly relevant and non-relevant resume thus; SBERT with Cosine Similarity is helpful.
- the model produced a consistent and explainable ranking performance, without involving manual feature engineering.

## 7.2 QUALITATIVE RESULTS

- The snippet extraction aspect identified the prime sentence(s) within each resume to accelerate understanding of the reasons for a resulting rank.
- Example: a sentence from the number one ranked resume read, “Engaged with end-to-end the data collection using qualitative and quantitative approaches for 10+ different research projects,” which directly affected analytical needs of the job description.
- The visualization of the outcomes on the interface had also made the matching process easier to interpret, and thus, helped build trust in the AI ranked resumes.
- Also, compatibility with multi-format (PDF, DOCX, TXT) resumes resulted in a seamless experience across the variety of resume formats used.

## 7.3 COMPARATIVE ANALYSIS

| Algorithm                   | Approach                    | Accuracy  | Interpretability |
|-----------------------------|-----------------------------|-----------|------------------|
| TF-IDF                      | Keyword-based matching      | Moderate  | Low              |
| Cosine Similarity           | Vector-based comparison     | High      | Medium           |
| Sentence-BERT (SBERT)       | Semantic contextual model   | Very High | High             |
| Sentence Snippet Extraction | Local semantic highlighting | High      | Very High        |

## 7.4 COMPARISON DISCUSSION

- The SBERT + Cosine Similarity pipeline outperformed the traditional TF-IDF model regarding contextual understanding and ranking certainty.
- The traditional TF-IDF approach considered only the overlaps of words, while the SBERT + Cosine Similarity pipeline captures semantically relevant phrases using a semantic distance, for e.g., The terms “data analysis” and “data insights” are semantically relevant.
- SBERT provides contextual meanings of the keywords and synonyms beyond basic relevancy metrics in TF-IDF.
- The phrase-based extraction algorithm afforded professionalism, something the traditional TF-IDF metric does not, prior to analyzing the snippet.

**RM Resume Matcher**  
Semantic search · Sentence-BERT embeddings · Elegantly presented results

**Job Description**  
Paste the job description here...

Support: .pdf · .docx · .txt 0 characters

**Ready to match**  
Paste job description and upload resumes to rank candidates.

**Upload Resumes**  
Choose files Choose Files No file chosen Match Resumes

Drag & drop files onto the Choose files button or click to open. Max file size depends on server settings.  
No files selected

**RM Resume Matcher**  
Semantic search · Sentence-BERT embeddings · Elegantly presented results

**Job Description**  
We are seeking a highly motivated and experienced Software Engineer to join our dynamic team. The ideal candidate will have a strong background in software development, excellent problem-solving skills, and the ability to work effectively in a collaborative environment.  
Key Responsibilities:  
- Develop, test, and maintain high-quality software applications.  
- Collaborate with cross-functional teams to define and design new features.  
- Troubleshoot and resolve software defects and issues.  
- Participate in code reviews to ensure adherence to coding standards and best practices.  
- Continuously learn and apply new technologies to improve software performance and scalability.  
Support: .pdf · .docx · .txt 1579 characters

**Top Match**  
data-analyst2\_-\_Template\_18\_15.pdf  
Summary: highest semantic similarity  
Growthsi, London, United Kingdom & Barcelona, Spain Career training and membership SaaS with 150,000 users Research Assistant 11/2018 – 09/2019 • Engaged in end-to-end data collection processes using qualitative and quantitative methods for 10+ research projects.  
38.51% Rank #1

**Upload Resumes**  
Choose files Choose Files No file chosen Match Resumes

Drag & drop files onto the Choose files button or click to open. Max file size depends on server settings.  
No files selected

**Matching results:**

| # | Filename                                                   | Snippet | Match |
|---|------------------------------------------------------------|---------|-------|
|   | Growthsi, London, United Kingdom & Barcelona, Spain Career |         |       |

| All Uploaded Resumes (ranked) |                                    | 4 files                                                                                                                                                                                                                                                                            |        |
|-------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| #                             | Filename                           | Snippet                                                                                                                                                                                                                                                                            | Match  |
| 1                             | data-analyst2_-_Template_18_15.pdf | <p>Growthsi, London, United Kingdom &amp; Barcelona, Spain Career training and membership SaaS with 150,000 users Research Assistant 11/2018 – 09/2019 • Engaged in end-to-end data collection processes using qualitative and quantitative methods for 10+ research projects.</p> | 38.51% |

|   |                                      |                                                                                                                                                                                                                                                                                                        |        |
|---|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 2 | data-scientist2_-_Template_18_14.pdf | <p>Growthsi, United Kingdom &amp; Barcelona, Spain Career training and membership SaaS with 150,000 users Database Developer 11/2018 – 09/2019 • Designed a database to track purchase orders, invoices, and shipping information from 1350+ clients.</p>                                              | 28.74% |
| 3 | Data_Analyst_-_ats_compatible_14.pdf | <p>Developed x that led to y% improvement<br/>EDUCATION<br/>Resume Worded University , San Francisco, CA 2012 Bachelor of Business Management, Minor in Data Analytics • Awards: Resume Worded Teaching Fellow (only 5 awarded to class), Dean's List 2012 (Top 10%) • Completed one-year study...</p> | 28.36% |
| 4 | data-scientist-resume-example_19.pdf | <p>Statistics Rutgers University September 2011 – April 2015 New Brunswick, NJ S K I L L S Python (NumPy, Pandas, Scikit-learn, Keras, Flask) SQL (MySQL, Postgres) Git Time Series Forecasting Productionizing Models Recommendation Engines Customer Segmentation AWSW O R K E X P E R...</p>        | 17.07% |

## **8. CONCLUSION & FUTURE WORK**

### **8.1 CONCLUSION**

The AI-based Resume and Job Description Matching System is a promising application that uses the Sentence-BERT embeddings and cosine similarity to improve recruitment in an automated manner. Its ability to convert text into semantics vectors allows the program to evaluate context relevance between resumes and job descriptions—a feat much greater than typical keyword match. The introduction of snippet generation enhances interpretability, so that HR professionals can immediately find critical evidence regarding a candidate's fitness. The inclusion of hybrid methodologies (SBERT+ TF-IDF) ensures depth and robustness, while the web interface's Flask architecture offers a convenient environment for real-time ranking and analysis. At its core, the project is a proof-of-concept for how NLP and transformer based models can totally transform candidate screening into something that's faster, more intelligent, and fair.

### **8.2 FURTHER WORK**

- Implement Named Entity Recognition to extract entities including skills, organizations, and experience duration.
- Develop a feedback loop to learn from recruiter preferences.
- Integrate with cloud-based APIs for large-scale deployment.
- Include data visualization dashboards for HR analytics.
- Add multilingual and domain-adaptive models for global recruitment.

## **REFERENCES:**

- Applying BERT-Based NLP for Automated Resume Screening and Candidate Ranking:  
<https://www.researchgate.net/publication/378829570> [Applying BERT-Based NLP for Automated Resume Screening and Candidate Ranking](#)
- Resume2Vec: Transforming Applicant Tracking Systems with Intelligent Resume Embeddings:  
<https://www.mdpi.com/2079-9292/14/4/794>
- Fine-tuned Siamese Sentence-BERT for Matching Jobs and Vacancies:  
<https://arxiv.org/abs/2109.06501>
- Enhanced Resume Screening for Smart Hiring Using Sentence-BERT:  
[https://thesai.org/Downloads/Volume15No8/Paper\\_25-Enhanced Resume Screening for Smart Hiring.pdf](https://thesai.org/Downloads/Volume15No8/Paper_25-Enhanced_Resume_Screening_for_Smart_Hiring.pdf)
- Automatic Resume Screening Using SBERT  
<https://www.ijfmr.com/research-paper.php?id=57538>